Appendix to the Group Project Report

Team 5 · Q15 — What should the "you may also like" shelf show, and to whom?

2026-11-26

Team number	Team 5
Members	Nini Rizhamadze, Giulia Scarpa, Milena Moschetti, Thi Thuy Khuc, Lina-Sophie Reinsch, Hai Yen Bui
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6. Ethics & compliance implications

Stake. Algorithmic bias in the recommender: a history-trained shelf is 93% written by heavy buyers, so almost half of buyers (light / occasional) would be served a shelf that does not reflect their behaviour.

Who bears it. First-time and light buyers see recommendations shaped by customers who bought 6–20 times in the quarter. Feedback-loop risk (untested): once live, the shelf would create the pairs it later mines.

Misinformation. Labelling a history-based shelf “frequently bought together” would be false for the top history pairs (same-order lift < 1). Safeguard: label the shelf by what it shows, e.g. “often bought in the same order”.

Privacy. A history shelf profiles each customer’s full purchase record; an order shelf needs only the current basket (PIPL data-minimisation). Under China’s 2022 Algorithmic Recommendation Provisions (Art. 17), the store must offer an option not based on the user’s profile and an easy way to switch recommendations off.

Safeguard tied to the recommendation. Mine future rules only on the control half of the A/B test, report add-on rates by buyer type, and drop the action under the conditions in §5 of the main report.

7. Artificial intelligence (AI) use appendix

We used **Claude (Anthropic)** and **Cursor (Composer)** to help draft and revise notebook code, the report outline, the final report text, and the pitch slides; to propose robustness checks (first-order slice, heavy-buyer cut-off table, A/B kill conditions, shelf decision cards); and to format tables and figures. The team reviewed every cell and every number against the

notebook outputs from the official Lumen Beauty course dataset. No platform personal data left the course environment.

Every member of this team can explain every line of code and every number in this submission.

A. Supporting detail for the findings

Support dial on order baskets: at 5% / 2% / 1% product support → 0 rules; at 0.4% → 30 rules, all with lift > 1 (lowest 1.39). Largest product-pair support = 0.79%. History mining at 2% returns 870 rules. Serum-without-cream orders by month: Sep 137, Oct 124, Nov 121. Average cream line value ¥347. Scenario at 5% / 10% / 15% conversion: ¥2,206 / ¥4,412 / ¥6,619 per month.